MILITARY SPECIFICATION SHEET

CABLE, ELECTRIC, FILTER LINE, UNSHIELDED, UNJACKETED, MULTIPLE-COMPONENT, RADIO FREQUENCY ABSORPTIVE, 150°C, 600-VOLT

This specification sheet is approved for use by all departments and agencies of the Department of Defense.

The complete requirements for procuring the cable described herein shall consist of this document and the issue in effect of Specification MIL-C-85485.

Component Wires -
M85485/5 for Tin-Coated Copper
M85485/6 for Silver-Coated High-Strength Copper Alloy

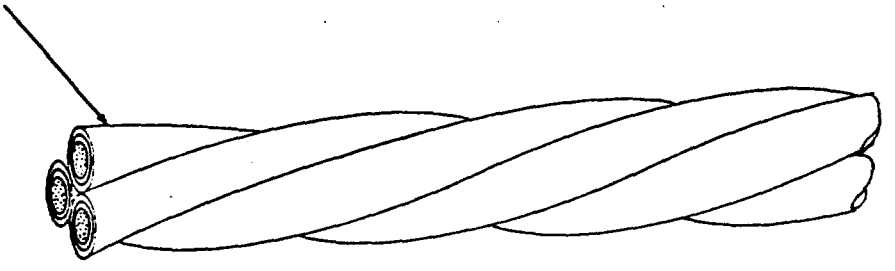


FIGURE 1. Typical configuration.

Ratings:

TEMPERATURE RATING: 150°C (302°F) max continuous conductor temperature
VOLTAGE RATING: 600 volts (rms) at sea level

Additional requirements:

ATTENUATION (INSERTION LOSS): The cable of this specification sheet meets the attenuation requirements for the applicable wire size and number of components when shielded in accordance with M85485/8-*****. Consult QPL source for details.

INSULATION FLAWS: Spark Test - 3.0 kilovolts (rms), 60 Hz
Impulse Dielectric Test - 8.0 kilovolts (peak)

THERMAL STABILITY: 1250 hours at 180° ±3°C (356° ±5°F). The cable of this specification sheet, when shielded in accordance with M85485/8-*****, shall meet the voltage withstand (dielectric) of 1500 volts (rms) 60 Hz and the stop band attenuation requirements for the applicable component wire size and number of components. Consult QPL source for details.

VOLTAGE WITHSTAND (DIELECTRIC) Conductor to conductor 1500 volts (rms) 60 Hz

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TABLE I. Construction details.

Part Number 1/	Conductor Size	Number of Components	Outside Diameter (in.)		Maximum Weight (lbs/1000 ft.)
			(nom)	(max)	
M85485/7-24*2†	24	2	.112	.118	8.0
M85485/7-22*2†	22	2	.124	.130	10.4
M85485/7-20*2†	20	2	.140	.147	14.3
M85485/7-18*2†	18	2	.160	.168	19.8
M85485/7-16*2†	16	2	.172	.181	23.9
M85485/7-14*2†	14	2	.200	.210	33.9
M85485/7-24*3†	24	3	.121	.127	11.9
M85485/7-22*3†	22	3	.134	.141	15.6
M85485/7-20*3†	20	3	.151	.159	21.4
M85485/7-18*3†	18	3	.173	.182	29.7
M85485/7-16*3†	16	3	.186	.195	35.8
M85485/7-14*3†	14	3	.216	.227	50.8
M85485/7-24*4†	24	4	.136	.162	15.9
M85485/7-22*4†	22	4	.150	.179	20.8
M85485/7-20*4†	20	4	.169	.202	28.6
M85485/7-18*4†	18	4	.194	.231	39.6
M85485/7-16*4†	16	4	.208	.248	47.7
M85485/7-14*4†	14	4	.242	.289	67.7
M85485/7-24*5†	24	5	.151	.176	19.9
M85485/7-22*5†	22	5	.167	.195	26.0
M85485/7-20*5†	20	5	.189	.221	35.7
M85485/7-18*5†	18	5	.216	.252	49.5
M85485/7-16*5†	16	5	.232	.244	59.7
M85485/7-14*5†	14	5	.270	.315	84.7

1/ PART NO.: The asterisks in the part number column shall be replaced by component wire material designators in accordance with MIL-C-85485. The daggers in the part number column shall be replaced by color code designators in accordance with MIL-C-85485.

Application Notes:

The component wires in these constructions are effective radio frequency absorptive wires only when used as components in a shielded cable.

The component wires of M85485/7 are compatible with all standard connector and contact types, except that the primary jacket diameters of M85485/5 - 22 component wires exceed the allowable diameter for MIL-C-38999 size 22D contacts. However, these component wires can be terminated to size 22D contacts if the primary jacket and filter layer are stripped in the grommet area, leaving the primary insulation intact.

Qualifying Activity: The activity responsible for this qualified products list is the Naval Air Systems Command, Washington, D.C. 20361.

Custodian:
AF - 85

Review activity:
AF - 19

Preparing activity:
Navy - AS
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